DAY 11 — Feature Engineering & Data Preprocessing for ML Models

Day 11 focused on **Feature Engineering**, one of the most impactful steps in improving model accuracy. Machine learning models rely on meaningful features, and today's learning helped transform raw data into high-quality input that ML algorithms can understand.

🔹 1. Understanding Feature Engineering

Feature engineering includes:

- Creating new columns
- Transforming raw features
- Encoding categorical variables
- Scaling numerical data
- Handling dates & timestamps

We learned why good feature engineering often matters more than the choice of model.

🔸 2. Creating New Features

We practiced generating useful columns like:

- Percentage changes
- Moving averages
- Rolling windows
- Cumulative sums
- Daily price differences

For stock prediction, moving averages (MA5, MA10) significantly improve trend understanding.

🔸 3. Handling Categorical Data

We used encoding techniques:

- Label encoding
- One-hot encoding
- Ordinal encoding

Even though stock datasets contain fewer categorical variables, this knowledge is essential for general ML workflows.

🔸 4. Normalization & Standardization

We practiced:

MinMaxScaler

StandardScaler

RobustScaler

Scaling helps:

- Speed up training
- Prevent large-value domination
- Improve convergence of ML algorithms

🔸 5. Date & Time Feature Extraction

From timestamps, we extracted:

- Year
- Month
- Day
- Week of year
- Day of week

These features were later extremely valuable for the Prophet forecasting model.

🔸 6. Handling Imbalanced Data

Although not heavily relevant to stock forecasting, we studied:

- Oversampling
- Undersampling
- SMOTE

These techniques help when working with classification datasets.

Key Takeaways from Day 11

- Good features = good machine learning results
- Scaling is crucial for distance-based algorithms

- Date-time features significantly improve time-series models
- Feature engineering is a creative and analytical process

Today’s session strengthened my ability to refine datasets intelligently before modeling.